

Drill Problems 10

*Author: Jaehoon Song**Release: 2025-07-22 16:24:06-04:00***Purpose and Usage:**

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C O L U M B I A A C A D E M Y

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1. **Budget Interpretation** (10 points)

Maximum Mail Order Company upgrades its computers each year. The function $P(t) = 36t + 410$ models the company's budget for upgrades, in thousands of dollars, t years after 2004. What is the best interpretation of 36 in this context?

- A. The number of years until the budget is completely spent
- B. The amount budgeted, in thousands of dollars, in the year 2004
- C. The annual increase, in thousands of dollars, in the budget each year after 2004
- D. The number of years in which the budget will increase after 2004

Answer:

□

2. **Algebraic Expression Equivalence** (10 points)

Which expression is equivalent to $\frac{4x^2-3}{2x+\sqrt{3}}$?

- A. $2x - 3$
- B. $2x - \sqrt{3}$
- C. $2x + \sqrt{3}$
- D. $2x + 3$

Answer:

□

3. **y -intercept of Exponential Equation** (10 points)

$$y - 7 = 3^x - 5$$

What is the y -intercept for the given equation?

Answer:

□

4. **Linear Model for Calf Weight** (10 points)

A linear model estimates that the mean weight of a calf in New Zealand, at birth, is 34 kilograms. At 180 days after birth, the mean weight is 272 kilograms. Which function best defines this model, where $f(x)$ is the mean weight of a calf, in kilograms, x days after it is born, for $0 \leq x \leq 180$?

- A. $f(x) = 0.76x + 34$
- B. $f(x) = 0.76x + 180$
- C. $f(x) = 1.32x + 34$
- D. $f(x) = 34x + 272$

Answer:

□

5. **Function Table Value** (10 points)

For the function f , the table shows some values of x and their corresponding values of $f(x)$. If $f(1) = t$,

x	$f(x)$
1	3
3	6
5	9
7	12
9	15

what is the value of $f(3t)$?

Answer:



6. **Percent Increase from Chart** (10 points)

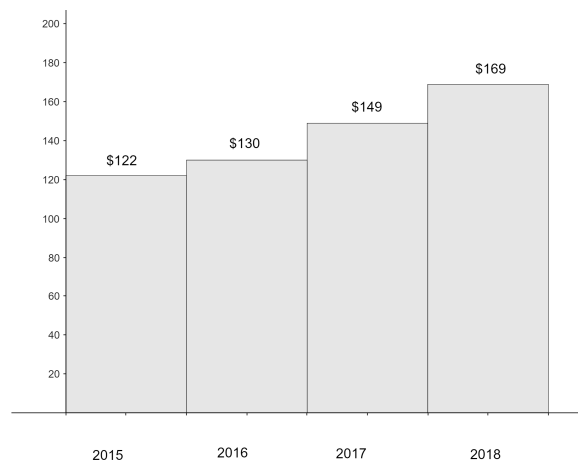


Figure 1: reference attached

The chart shows the average cost of housing, in dollars per square foot, in Nebraska from 2015 to 2018. By approximately what percent did the average cost of housing increase over this period?

- A. 20%
- B. 28%
- C. 39%
- D. 47%

Answer:



7. **Bacteria Growth Interpretation** (10 points)

$$p(t) = 2034(3)^{\frac{t}{14}}$$

The given function shows the number of bacteria in a pond after t days since the bacteria began to populate the pond. Which of the following is the best interpretation of $(3)^{\frac{t}{14}}$ in the equation?

- A. The number of bacteria at the beginning of the study
- B. The number of bacteria triples every two weeks
- C. The number of bacteria increases by 3 every two weeks
- D. The number of bacteria in the pond after two weeks

Answer:



8. **Factoring Polynomial** (10 points)

$$12x^4 - 8x^3 - 15x^2$$

The given expression is equal to $x^2(qx + 5)(rx - 3)$, where $q > 0$ and $r > 0$. What is the value of $q + r$?

Answer:



9. **Equation of Parallel Line** (10 points)

Line a is parallel to line b . Line a goes through the points $(-4, 0)$ and $(0, 6)$. If line b goes through the point $(-6, 0)$, what is the equation for line b ?

- A. $y = \frac{2}{3}x - 6$
- B. $y = \frac{2}{3}x + 4$
- C. $y = \frac{3}{2}x + 6$
- D. $y = \frac{3}{2}x + 9$

Answer:



10. **Equilateral Triangle Exterior Angle** (10 points)

The exterior angle of an equilateral triangle is equal to $(8x - 20)^\circ$. What is the value of x ?

- A. 5
- B. 10
- C. 12.5
- D. 17.5

Answer:



11. **Quadratic Equation Solutions** (10 points)

$$(x - 1)(x + 2) = x(x - 3)$$

How many solutions does the given equation have?

- A. 0
- B. 1
- C. 2
- D. 4

Answer:



12. **System of Inequalities Solution** (10 points)

$$y > -2x - 1$$

$$3y < x + 9$$

Which of the following coordinates would be true for the given system of inequalities?

- A. $(-3, 3)$
- B. $(-2, 1)$
- C. $(1, 4)$
- D. $(3, 1)$

Answer:



13. **Infinite Solutions in System of Equations** (10 points)

$$9x - 14y = -3$$

$$ax - by = 6$$

If the given system of equations has infinite solutions, what is the value of b ?

Answer:



14. **Equivalent Forms of Quadratic Function** (10 points)

$$f(x) = 6x^2 - 45x + 75$$

Which of the following equivalent forms of the given function f displays the x -intercepts as constants or coefficients?

- A. $f(x) = (2x - 5)(3x - 15)$
- B. $f(x) = 3x(2x - 15) + 75$
- C. $f(x) = 6\left(x - \frac{15}{4}\right)^2 + \frac{75}{8}$
- D. $f(x) = 6(x - 5)\left(x - \frac{5}{2}\right)$

Answer:

□

15. **System with One Real Solution** (10 points)

$$\begin{aligned}y &= 4 \\ y &= x^2 + 12x + n\end{aligned}$$

In the given system of equations, n is a positive constant. For what value of n does the system have exactly one real solution?

Answer:

□

16. **Minimum Value in System of Inequalities** (10 points)

$$\begin{aligned}y &\geq 2(x - 3)^2 + 5 \\ y &\leq x + 3\end{aligned}$$

If (a, b) is a solution to the given system of inequalities, what is the minimum value of b ?

- A. -5
- B. -3
- C. 3
- D. 5

Answer:

□

17. **Bacteria Doubling Problem** (10 points)

Biology students grew a culture of bacteria at a constant temperature of 55°C for 3 days. The students began with 200 spores and the number of spores doubled every 12 hours. Of the equations given, which can be used to correctly calculate the number of spores in the bacteria culture at the end of 3 days?

- A. $y = 200(12)^3$
- B. $y = 200(2)^3$
- C. $y = 200(2)^6$
- D. $y = 200(12)^2$

Answer:



18. **Comparing Means and Standard Deviations** (10 points)

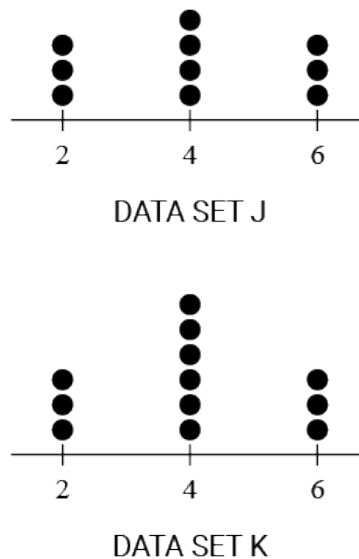


Figure 2: reference attached

Which of the statements best compares the means and standard deviations of Data Set J and Data Set K ?

- A. The mean of Data Set J is less than the mean of Data Set K , and the standard deviations of the two sets are equal.
- B. The mean of Data Set J is greater than the mean of Data Set K , and the standard deviations of the two sets are equal.
- C. The means of the two sets are equal, and the standard deviation of Data Set J is less than the standard deviation of Data Set K .
- D. The means of the two sets are equal, and the standard deviation of Data Set J is greater than the standard deviation of Data Set K .

Answer:



19. **Circumference of a Circle** (10 points)

What is the circumference of the circle with an equation of $x^2 + 6x + y^2 - 4y = 51$?

- A. 8π
- B. 14.28π
- C. 16π
- D. 64π

Answer:



20. **Trigonometric Value in Triangle** (10 points)

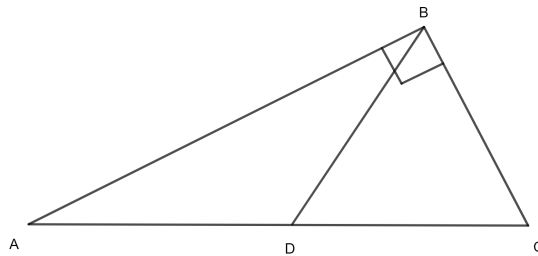


Figure 3: reference attached

In the triangle shown, $\angle ABC = 90^\circ$. Line segment BD is shown with point D lying on line AC . What is the value of $\sin \angle ABD - \cos \angle DBC$?

- A. 0
- B. 30
- C. 45
- D. 90

Answer:



21. **No Solution for System of Equations** (10 points)

$$y = (x - 6)(x + 6)$$
$$y = k$$

In the system of equations shown, k is an integer constant. What is the largest value of k for which the system has no solution?

Answer:



22. **Exponential Function Shift** (10 points)

The base of the exponential function $g(x)$ is k , where k is a positive real number, and k is equivalent to the value of the y -coordinate of the y -intercept of the graph of $g(x)$. Which of the following functions describes the exponential function $g(x - 1)$?

A. $g(x - 1) = k^x$

B. $g(x - 1) = k^{x+1}$

C. $g(x - 1) = k^x + k$

D. $g(x - 1) = k^{x-1} + k$

Answer:

